



— LAUGHTALE · CCNA 200-301 V1.1 · PRINTABLE PACK

Packet Tracer *Lab* *Scenarios*

Five progressive builds with topology tables, task lists, verification checklists and deliberate break-and-fix work.

5

Labs

2×

Builds each

45–60

Min cold

LAB 1 · SWITCHED ACCESS

LAB 2 · STP & INTER-VLAN

LAB 3 · ROUTING

LAB 4-5 · SERVICES, SECURITY & WLAN

- Build the topology yourself — no pre-built files
- A task is not complete until its verification command proves it

PRINT · A4 PORTRAIT · SINGLE COLUMN · BACKGROUND GRAPHICS ON

Overview

Part of the CCNA 200-301 v1.1 Study Master Plan — see 00_Index_and_How_To_Use.md for notation and 02_Study_Calendar.md for when to build each lab.

Section 3 — Packet Tracer Lab Scenarios

Lab Rules

1. Build the topology yourself. Do not download a pre-built file. Placing and cabling the devices is part of the exercise.
2. No CLI answers are given below, by design. Task lists state requirements, not commands. If you cannot produce the command, that is the finding — go back to the video, then return.
3. No ? and no Tab completion on the second and subsequent attempts at any lab.
4. Verify before you move on. A task is not complete until its verification command proves it.
5. Save your work before breaking anything, then break it deliberately. Fault injection is where the real learning happens — each lab ends with a break-and-fix section.
6. Build each lab twice: once with notes (Phase 1), once cold (Week 10). Addressing convention used throughout (adapt freely, but stay consistent):
 - Management/loopbacks: 10.255.255.0/24
 - User VLANs: 192.168.<VLAN>.0/24
 - Router-to-router links: 10.0.<link>.0/30
 - IPv6: 2001:db8:<site>:<subnet>::/64

LAB 1 — SWITCHED ACCESS LAYER

COVERS 2.1, 2.2, 2.3, 2.4, 5.7 (PORT SECURITY), 1.13

BUILD AFTER WEEK 5

REBUILD COLD WEEK 10

TARGET TIME COLD 45 MINUTES

Topology Objective

DEVICE	ROLE	CONNECTIONS
SW1	Access switch	Two links to SW2 (for the EtherChannel), PC1, IP Phone + PC3
SW2	Access switch	Two links to SW1, one link to SW3, PC2
SW3	Access switch	One link to SW2, PC4
PC1, PC2, PC3, PC4	Endpoints	One each to SW1/SW2/SW1/SW3
IP Phone	Voice endpoint	SW1, with PC3 daisy-chained behind it

VLAN plan: VLAN 10 = Sales, VLAN 20 = Engineering, VLAN 30 = Voice, VLAN 99 = Native/Management, VLAN 999 = Parking (unused ports).

Task List

- ☐ T1.1 Set a unique hostname on each switch and disable DNS lookup so typos don't hang the console.
- ☐ T1.2 Create VLANs 10, 20, 30, 99, and 999 on all three switches with descriptive names.
- ☐ T1.3 Assign PC1 to VLAN 10, PC2 to VLAN 20, PC4 to VLAN 10. Configure the ports as static access ports (do not leave them negotiating).
- ☐ T1.4 Configure the SW1 port serving the IP Phone as an access port in VLAN 20 for the daisy-chained PC3, with VLAN 30 carrying voice traffic on the same physical port.
- ☐ T1.5 Bundle the two SW1–SW2 links into a Layer 2 LACP EtherChannel. SW1 must initiate actively; SW2 must respond without initiating. Configure all trunking parameters on the logical interface, not the members.
- ☐ T1.6 Configure the SW2–SW3 link as a static 802.1Q trunk.
- ☐ T1.7 Change the native VLAN on all trunks to VLAN 99. Confirm the change is symmetric on both ends of every trunk.
- ☐ T1.8 Prune the trunks so only VLANs 10, 20, 30, and 99 are permitted. VLAN 999 must not cross any trunk.
- ☐ T1.9 Move every unused switchport into VLAN 999 and administratively shut them down.
- ☐ T1.10 Enable port security on the PC1 and PC2 access ports: maximum two MAC addresses, sticky learning, violation mode restrict.
- ☐ T1.11 Enable port security on the PC4 access port with maximum one MAC address and violation mode shutdown.
- ☐ T1.12 Ensure CDP is running, then additionally enable LLDP globally — remembering that LLDP transmit and receive are separate settings.
- ☐ T1.13 Give each switch a management IP address in VLAN 99 and a default gateway.

Verification Checklist

- ☐ show vlan brief on all three switches — every VLAN exists, every access port is in the right one
- ☐ show interfaces trunk on all three — correct trunk ports, correct native VLAN, correct allowed and active VLAN lists
- ☐ show etherchannel summary — port-channel shows (SU) and both members show (P)
- ☐ show interfaces port-channel <n> — logical interface up
- ☐ show interfaces switchport on the phone port — confirms access VLAN 20 and voice VLAN 30
- ☐ show port-security and show port-security interface <int> — correct max, correct violation mode, sticky addresses learned
- ☐ show port-security address — sticky entries present
- ☐ show running-config interface <int> — sticky MACs written into the config
- ☐ show cdp neighbors detail and show lldp neighbors detail — full neighbour list, correct port mapping
- ☐ show mac address-table — endpoints learned in the correct VLANs
- ☐ ping PC1 → PC4 (same VLAN, across all three switches) — must succeed
- ☐ ping PC1 → PC2 (different VLANs, no routing yet) — must fail. If it succeeds, your VLAN segmentation is broken.

Break and Fix

- ☐ Change the native VLAN on one end of a trunk only. Find and fix it using show interfaces trunk alone.
- ☐ Remove VLAN 10 from SW2 (the transit switch) and observe PC1↔PC4 break. Diagnose before fixing.
- ☐ Set one EtherChannel member port to a different mode or speed. Observe the flag change in show etherchannel summary.
- ☐ Connect a second MAC to the PC4 port and observe the err-disable. Recover the port.

LAB 2 — SPANNING TREE & INTER-VLAN ROUTING

COVERS 2.5, 2.1.C, 3.1, 3.2

BUILD AFTER WEEK 6

REBUILD COLD WEEK 10

TARGET TIME COLD 50 MINUTES

Topology Objective

DEVICE	ROLE	CONNECTIONS
DSW1	Layer 3 distribution switch	Trunks to ASW1, ASW2, and to DSW2
DSW2	Layer 3 distribution switch	Trunks to ASW1, ASW2, and to DSW1
ASW1	Access switch	Trunks to DSW1 and DSW2; PC1
ASW2	Access switch	Trunks to DSW1 and DSW2; PC2
R1	Router (router-on-a-stick)	Single trunk link to ASW1
PC3	Endpoint behind R1's routed segment	—

This is a full triangle-plus-square topology with deliberate physical loops. STP must block something — your job is to control what. VLAN plan: VLAN 10, VLAN 20, VLAN 99 (native/management).

Task List

- ☐ T2.1 Create VLANs 10, 20, and 99 on all four switches. Trunk every switch-to-switch link with native VLAN 99.
- ☐ T2.2 Set the spanning-tree mode to Rapid PVST+ on all four switches.
- ☐ T2.3 Make DSW1 the root bridge for VLAN 10 and the secondary/backup root for VLAN 20. Make DSW2 the root bridge for VLAN 20 and secondary for VLAN 10. Achieve this by manipulating priority — and be able to state the resulting numeric priority value for each VLAN.
- ☐ T2.4 Before verifying, predict on paper which port on each switch will be root, designated, or alternate for each VLAN. Then check whether you were right.
- ☐ T2.5 Enable PortFast on all access ports connecting to PCs. Do not enable it on any trunk.
- ☐ T2.6 Enable BPDU guard on those same access ports.
- ☐ T2.7 Enable root guard on the DSW1 and DSW2 ports facing the access switches, so no access switch can ever become root.
- ☐ T2.8 Configure router-on-a-stick on R1 for VLANs 10 and 20 using subinterfaces, with the correct 802.1Q encapsulation on each and the native VLAN handled correctly. Assign each subinterface the gateway address for its VLAN.
- ☐ T2.9 Verify inter-VLAN routing works through R1. Confirm PC1 can reach PC2.
- ☐ T2.10 Now migrate the routing: enable IP routing on DSW1 and create SVIs for VLANs 10 and 20, taking over the gateway addresses. Remove or shut down R1's role.
- ☐ T2.11 Make sure the SVIs come up — an SVI requires at least one active access port in the VLAN, or the VLAN to be active on a trunk.
- ☐ T2.12 Configure a routed port (or an SVI in a transit VLAN) between DSW1 and DSW2 and add a static or default route so both distribution switches have full reachability.
- ☐ T2.13 Compare the two approaches in writing: when is router-on-a-stick the right answer, and when is an L3 switch?

Verification Checklist

- ☐ show spanning-tree vlan 10 and show spanning-tree vlan 20 on every switch — confirm the intended root for each VLAN
- ☐ show spanning-tree root — root bridge, root cost, and root port per VLAN
- ☐ show spanning-tree summary — mode is Rapid PVST+, PortFast and BPDU guard status
- ☐ show spanning-tree interface <int> detail — port role, state, cost, and link type
- ☐ Confirm the blocked (alternate) port differs between VLAN 10 and VLAN 20. If it doesn't, your priority tuning didn't take effect — that's the whole point of per-VLAN spanning tree.
- ☐ show ip route on DSW1 and DSW2 — connected and local routes for each SVI, plus the transit route
- ☐ show ip interface brief — all SVIs up/up
- ☐ show vlan brief — VLANs active so the SVIs stay up
- ☐ ping PC1 → PC2 (inter-VLAN) — must succeed
- ☐ tracer PC1 → PC2 — confirms which device is actually doing the routing after the migration
- ☐ ping PC1 → each SVI gateway — must succeed

Break and Fix

- ☐ Shut the forwarding uplink on ASW1 and watch RSTP converge. Time it. Compare that convergence to what legacy STP would have taken.
- ☐ Plug a switch into a BPDU-guard-protected PortFast port. Observe the err-disable and recover it.

- ☐ Lower an access switch's priority below the root's and watch root guard put the DSW port into root-inconsistent state. Then fix it.
- ☐ Omit the encapsulation command on one R1 subinterface and diagnose why that VLAN loses its gateway.

LAB 3 — IP CONNECTIVITY & DYNAMIC ROUTING

COVERS 3.1, 3.2, 3.3, 3.4, 1.8, 1.9

BUILD AFTER WEEK 7

REBUILD COLD WEEK 10

TARGET TIME COLD 55 MINUTES

Topology Objective

DEVICE	ROLE	CONNECTIONS
R1	Edge router	R2 (primary link), R3 (backup link), LAN1
R2	Core router	R1, R3, R4, LAN2
R3	Core router	R1, R2, R4
R4	Remote router	R2, R3, LAN3
SW1-SW3	One access switch per LAN	—
PC1-PC3	One endpoint per LAN	—

Four routers in a partial mesh with redundant paths. Every router gets a loopback interface used as its OSPF router ID and management address. Run IPv4 and IPv6 in parallel (dual stack) throughout.

Task List Part A — Addressing and static routing

- ☐ T3.1 Address every router-to-router link from a /30 scheme you design yourself, and every LAN from your /24 scheme. Document the plan before typing anything.
- ☐ T3.2 Create a loopback on each router: R1 10.255.255.1/32, R2 .2, R3 .3, R4 .4.
- ☐ T3.3 Enable IPv6 routing globally and dual-stack every interface with a global unicast address. Use EUI-64 on at least two interfaces and static addressing on the rest, so you have practised both.
- ☐ T3.4 Before any dynamic routing: configure IPv4 static routes on R1 so it reaches LAN3 via R2. Use a fully-specified static route (both exit interface and next-hop).
- ☐ T3.5 Configure a floating static route on R1 for the same destination via R3, with an administrative distance high enough that it only installs when the primary path fails but low enough that it beats nothing else. Test it by shutting the R1–R2 link.
- ☐ T3.6 Configure a default route on R4 pointing toward the core, and confirm the gateway of last resort appears.
- ☐ T3.7 Configure a host route (/32) on R2 to R4's loopback and observe how longest-prefix match selects it over any broader route.
- ☐ T3.8 Configure IPv6 static routes including a default route, and at least one route using a link-local next hop — which requires specifying the exit interface.
- ☐ T3.9 Remove the static routes for LAN-to-LAN reachability (keep the floating static as a deliberate backup) before moving to Part B. Part B — Single-area OSPFv2
- ☐ T3.10 Enable OSPF process 1 on all four routers, area 0 only.
- ☐ T3.11 Set each router's router ID manually to its loopback address. Verify the RID actually changed — remember what is required for a RID change to take effect.
- ☐ T3.12 Advertise all router-to-router links and all LANs into OSPF using wildcard masks (not subnet masks). Also advertise the loopbacks.
- ☐ T3.13 Configure passive interfaces on every LAN-facing interface and on the loopbacks, so no hellos are sent where there are no OSPF neighbours — while keeping those subnets advertised. Use the default-passive approach with explicit exceptions.
- ☐ T3.14 On the R2–R3 link, manipulate OSPF interface priority so that R2 becomes the DR and R3 the BDR on that segment. Then prove the election is non-preemptive by setting R3's priority higher after the adjacency forms and confirming nothing changes until you clear the process.
- ☐ T3.15 On the R1–R2 link, change the OSPF network type to point-to-point and confirm that no DR/BDR is elected on it.
- ☐ T3.16 Change the reference bandwidth on all four routers so gigabit and faster links have distinct costs. Do it on every router — a mismatch causes suboptimal routing.
- ☐ T3.17 Manually set the OSPF cost on one interface to steer traffic down a specific path. Prove the path changed with a traceroute.

- ☐ T3.18 On R1, configure a default route to a simulated internet (a loopback standing in for the ISP) and inject it into OSPF, so R2, R3, and R4 all learn it.

Verification Checklist

- ☐ show ip interface brief and show ipv6 interface brief — every interface up/up with the right address
- ☐ show ip ospf neighbor — every expected adjacency in FULL state, with correct DR/BDR/DROTHER roles
- ☐ show ip ospf interface <int> — network type, cost, hello/dead timers, priority, DR/BDR identity
- ☐ show ip ospf interface brief — quick sweep of all OSPF-enabled interfaces
- ☐ show ip protocols — router ID, passive interfaces listed, networks advertised
- ☐ show ip route ospf — OSPF routes present with AD 110, plausible metrics
- ☐ show ip route — confirm the gateway of last resort is set, and identify every route code present
- ☐ show ip route <specific prefix> — walk through why this route was chosen (longest match → AD → metric)
- ☐ show ipv6 route — IPv6 statics and connected routes present
- ☐ ping PC1 → PC3 and PC2 → PC3 — must succeed
- ☐ ping IPv6 between routers, including a link-local ping specifying the exit interface
- ☐ tracer PC1 → PC3 — path matches what your cost manipulation intended

Break and Fix Break OSPF five separate ways and diagnose each using only show commands before fixing:

- ☐ Mismatched hello/dead timers on one link
- ☐ Mismatched subnet mask on one end of a link
- ☐ A passive interface where a neighbour actually exists
- ☐ Two routers configured with the same router ID
- ☐ An MTU mismatch (watch the adjacency hang in ExStart/Exchange — learn to recognise this signature)

Then: shut the R1–R2 link and confirm the floating static installs. Bring it back and confirm OSPF reclaims the path.

LAB 4 — FIRST-HOP REDUNDANCY & IP SERVICES

COVERS 3.5, 4.2, 4.3, 4.4, 4.5, 4.6, 4.8, 2.8, 5.3

BUILD AFTER WEEK 8

REBUILD COLD WEEK 10

TARGET TIME COLD 45 MINUTES

Topology Objective

DEVICE	ROLE	CONNECTIONS
R1	Gateway A (HSRP)	SW1, and to the core/server segment
R2	Gateway B (HSRP)	SW1, and to the core/server segment
SW1	Access switch	R1, R2, PC1, PC2
SRV1	DHCP / NTP / Syslog / TFTP server	On a different subnet from the clients — the relay is the point
PC1, PC2	DHCP clients	SW1

R1 and R2 both connect to the same client VLAN, providing a redundant default gateway.

Task List Part A — First-hop redundancy

- ☐ T4.1 Address R1 and R2 on the client VLAN with distinct real IPs from the same subnet.
- ☐ T4.2 Configure HSRP group 1 on both routers sharing a virtual IP that will become the clients' default gateway.
- ☐ T4.3 Make R1 the active router by priority, and enable preemption so R1 reclaims the active role after a failure — remembering that HSRP preemption is off by default.
- ☐ T4.4 Identify the virtual MAC address HSRP generates and confirm it matches the expected format for the group number.
- ☐ T4.5 Create a second VLAN and a second HSRP group where R2 is active and R1 standby, so the two routers load-share across VLANs rather than leaving one idle.

Part B — IP services

- ☐ T4.6 Configure R1 as an NTP master with a stratum you choose. Configure R2 and SW1 as NTP clients pointing at R1. Set the correct timezone on all devices.
- ☐ T4.7 Configure a DHCP pool on SRV1's router (or on the server device) for the client VLAN, handing out address, mask, default gateway (the HSRP virtual IP, not a real interface IP — this is the point of the exercise), DNS server, and domain name. Exclude the first ten addresses in the range.
- ☐ T4.8 Configure a DHCP relay agent on the client-VLAN interface of R1 and R2 pointing at the server. Then explain in writing why the relay is required.
- ☐ T4.9 Set PC1 and PC2 to obtain addresses via DHCP. Confirm they receive the correct gateway.
- ☐ T4.10 Configure syslog on all network devices to send to SRV1, at severity level 4 (warning). Then state exactly which severity levels will and will not be sent, and why.
- ☐ T4.11 Configure logging buffered locally as well, and set the buffer size.
- ☐ T4.12 Configure SSH on R1, R2, and SW1: hostname, domain name, RSA keys of at least 1024 bits, a local user with a privilege level and secret, VTY lines set to local login accepting SSH only, version 2 forced, and an exec-timeout.
- ☐ T4.13 Set enable secret on all devices. Apply service password-encryption, then state why it is not real security.
- ☐ T4.14 Apply a VTY access-class so only the management subnet may SSH to the devices.
- ☐ T4.15 Configure a login banner (MOTD) with an appropriate legal warning — and know why "Welcome" is the wrong banner text.

Verification Checklist

- ☐ show standby and show standby brief — group number, state (Active/Standby), priority, preemption enabled, virtual IP, virtual MAC
- ☐ Confirm R1 is active for VLAN A and R2 is active for VLAN B
- ☐ show ntp status — synchronised, correct stratum
- ☐ show ntp associations — the * marks the selected server
- ☐ show clock on every device — times match
- ☐ show ip dhcp binding — leases issued to PC1 and PC2
- ☐ show ip dhcp pool — correct range, excluded addresses, leases in use
- ☐ ipconfig /all on PC1 — address, mask, gateway is the HSRP virtual IP, DNS, and the DHCP server address
- ☐ show logging — destination, severity level, buffered messages
- ☐ show ip ssh — SSH enabled, version 2
- ☐ show ssh / show users — active sessions
- ☐ show running-config | section line vty — transport input, login local, access-class, exec-timeout
- ☐ Actually SSH from one device to another and confirm Telnet is refused
- ☐ ping PC1 → SRV1 — must succeed

Break and Fix

- ☐ Shut R1's client-VLAN interface. Confirm R2 takes over as active and that PC1 keeps pinging with at most a brief interruption. Bring R1 back and confirm preemption returns it to active.
- ☐ Remove the ip helper-address and observe DHCP fail. Diagnose from the client side first, then the router.
- ☐ Set the DHCP pool's default-router to R1's real IP instead of the virtual IP. Fail R1 and observe that clients now lose connectivity — which demonstrates exactly why the virtual IP must be the gateway.
- ☐ Set syslog severity to 7 and observe the volume difference.
- ☐ Attempt SSH from outside the management subnet and confirm the access-class blocks it.

LAB 5 — SECURITY & INFRASTRUCTURE PROTECTION

COVERS 5.6, 5.7, 5.10, 4.1, 1.11, 2.6, 2.7, 2.9

BUILD AFTER WEEK 8

REBUILD COLD WEEK 10

TARGET TIME COLD 55 MINUTES

Topology Objective

DEVICE	ROLE	CONNECTIONS
R1	Edge router — NAT/PAT boundary + ACL enforcement	Inside LAN, and outside link to ISP
ISP	Simulated internet	R1; plus a loopback or server acting as a public host
SW1	Access switch	R1, PC1 (Sales), PC2 (Engineering), SRV1 (internal server), rogue DHCP device, WLC/AP
SRV1	Internal server (HTTP + DNS + DHCP)	SW1
WLC + LAP	Wireless controller and lightweight AP	SW1
Laptop	Wireless client	—
Rogue-DHCP	An extra router/server set up as an unauthorised DHCP server	SW1

VLAN plan: VLAN 10 Sales, VLAN 20 Engineering, VLAN 30 Servers, VLAN 40 Wireless, VLAN 99 Management.

Task List Part A — NAT/PAT

- ☐ T5.1 Address the inside LAN from RFC 1918 space and the R1–ISP link from a public-looking range. Mark R1's inside and outside interfaces correctly for NAT.
- ☐ T5.2 Configure a static NAT mapping so SRV1 is reachable from the internet on a fixed public address.
- ☐ T5.3 Configure dynamic NAT with a pool (a small pool — two or three addresses) plus an ACL identifying the inside source addresses. Generate more simultaneous sessions than the pool can hold and observe what happens.
- ☐ T5.4 Reconfigure to PAT overloading the outside interface address. Confirm many inside hosts now share one public address, distinguished by port.
- ☐ T5.5 From the translation table, fill in a four-column chart by hand: inside local, inside global, outside local, outside global — for one specific session. Part B — ACLs
- ☐ T5.6 Write a standard numbered ACL that blocks the Sales VLAN from reaching SRV1 while permitting everything else. Place it correctly — remember standard ACLs go near the destination — and justify your placement in writing.
- ☐ T5.7 Replace it with a named standard ACL and demonstrate inserting a new permit statement in the middle using sequence numbers, without rebuilding the list.
- ☐ T5.8 Write an extended ACL that: permits Engineering to reach SRV1 on HTTP and HTTPS only; permits ICMP from anywhere to SRV1; denies all other traffic to SRV1; permits everything else. Place it near the source.
- ☐ T5.9 Write a named extended ACL applied outbound on R1's ISP-facing interface that blocks Telnet leaving the network but permits SSH.
- ☐ T5.10 Apply an ACL to R1's VTY lines so only the management VLAN may attempt SSH.
- ☐ T5.11 For each ACL, state in writing what the implicit deny at the end will catch, and prove it with a test that should fail.

Part C — Layer 2 protection

- ☐ T5.12 Enable DHCP snooping globally and on VLANs 10, 20, and 40. Trust only the uplink toward the legitimate DHCP server and the switch-to-switch links. Set a rate limit on untrusted ports.
- ☐ T5.13 Power on the rogue DHCP server on an untrusted port. Confirm clients do not receive addresses from it.
- ☐ T5.14 Enable Dynamic ARP Inspection on the same VLANs, with the same trusted ports. Note the ordering dependency — explain in writing why DAI cannot work without DHCP snooping's binding table first.
- ☐ T5.15 Add an ARP ACL for a statically-addressed host that has no DHCP snooping binding.
- ☐ T5.16 Re-apply port security on all access ports with sticky learning and an appropriate violation mode for each port type.

Part D — Wireless

- ☐ T5.17 Connect the LAP to SW1 and the WLC to SW1. Decide and justify: should the LAP port be access or trunk in this local-mode deployment? Should the WLC port be access or trunk? Configure accordingly.
- ☐ T5.18 Give the WLC a management interface address in VLAN 99 and a dynamic interface mapped to VLAN 40 for wireless clients.
- ☐ T5.19 Confirm the LAP joins the WLC. Identify which CAPWAP tunnels exist and what each carries.
- ☐ T5.20 In the WLC GUI, create a WLAN with your chosen SSID, map it to the VLAN 40 dynamic interface, and secure it with WPA2-Personal using AES and a pre-shared key.
- ☐ T5.21 Set the WLAN's QoS profile appropriately and enable the WLAN.

- ☐ T5.22 Associate the laptop, confirm it receives an address from the VLAN 40 DHCP scope, and confirm it can reach the internet through NAT.
- ☐ T5.23 Walk the WLC GUI tabs (General, Security, QoS, Advanced) and write down what each key setting does — this is objective 2.9's interpret skill.

Verification Checklist

- ☐ show ip nat translations — entries present; for PAT, confirm multiple inside locals share one inside global with different ports
- ☐ show ip nat statistics — hits, misses, pool utilisation, correct inside/outside interface list
- ☐ show ip interface <int> — confirms ip nat inside / ip nat outside and which ACLs are applied in which direction
- ☐ show access-lists — read the match counters. Zero matches on a rule you're testing means the traffic isn't reaching it
- ☐ show running-config | include access-class — VTY ACL applied
- ☐ show ip dhcp snooping — enabled VLANs, trusted interfaces, rate limits
- ☐ show ip dhcp snooping binding — binding table populated
- ☐ show ip arp inspection and show ip arp inspection statistics — enabled VLANs, drops recorded
- ☐ show port-security — all protected ports listed with correct modes
- ☐ show ip route on R1 — default route toward the ISP present
- ☐ WLC GUI: AP shows joined, WLAN shows enabled, client shows associated
- ☐ ping PC2 → SRV1 on permitted ports — must succeed
- ☐ ping/HTTP PC1 → SRV1 — must fail (standard ACL working)
- ☐ Telnet from inside to the ISP — must fail; SSH — must succeed
- ☐ ping PC1 → ISP public host — must succeed (NAT working)
- ☐ ipconfig on the laptop — VLAN 40 address, correct gateway
- ☐ ipconfig on a client with the rogue DHCP server running — must still hold the legitimate address

Break and Fix

- ☐ Reverse the ACL statement order so the deny precedes the permit it was supposed to follow. Observe the first-match failure and fix it.
 - ☐ Apply the extended ACL in the wrong direction on the interface. Diagnose using show ip interface plus match counters.
 - ☐ Remove ip nat inside from R1's LAN interface. Observe NAT silently stop working. This is the single most common real-world NAT fault.
 - ☐ Mark the rogue DHCP server's port as trusted and watch clients get poisoned. Then untrust it.
 - ☐ Enable DAI before DHCP snooping has built any bindings and observe legitimate ARP being dropped.
 - ☐ Put the WLC port in the wrong VLAN and diagnose why the AP never joins.
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Final Lab Audit

Before booking the exam, all of the following must be true.

- ☐ All five labs built from scratch at least twice
 - ☐ All five labs completed cold — no notes, no ?, no Tab completion
 - ☐ Every break-and-fix scenario diagnosed using show commands before fixing
 - ☐ Lab 1 cold in ≤ 45 min · Lab 2 ≤ 50 min · Lab 3 ≤ 55 min · Lab 4 ≤ 45 min · Lab 5 ≤ 55 min
 - ☐ Every show command in every verification checklist can be typed from memory and its output interpreted
 - ☐ All 53 blueprint objectives in Section 1 have three ticks
 - ☐ Practice exam average above 85% across three separate attempts
 - ☐ Subnetting: any problem solved in ≤ 30 seconds
 - ☐ Weak Point Register fully resolved
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End of plan. Print Section 1 and tick as you go — the checkboxes are the progress tracker, not a decoration.